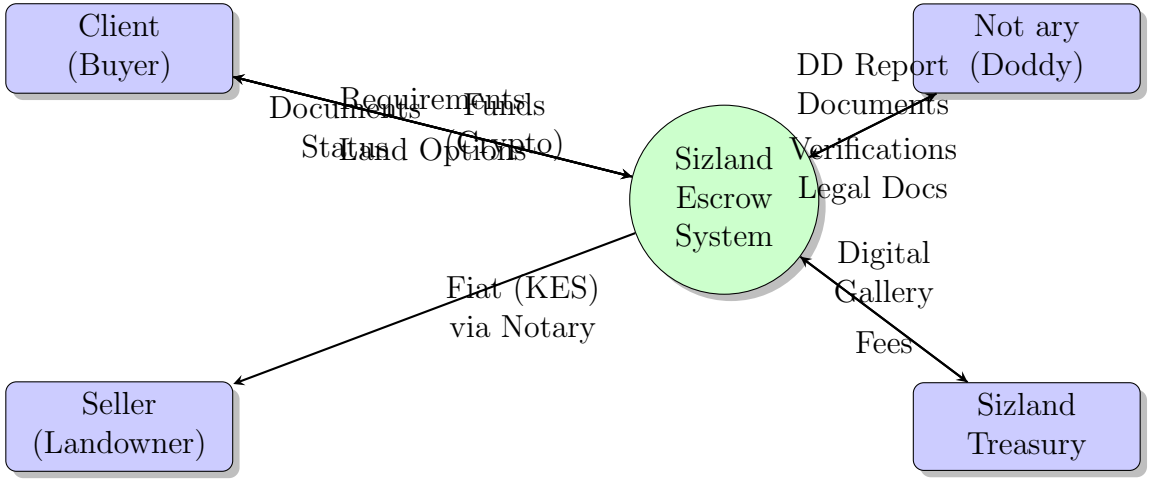
Sizland Land Acquisition Data Flow Diagram

Complete System Architecture

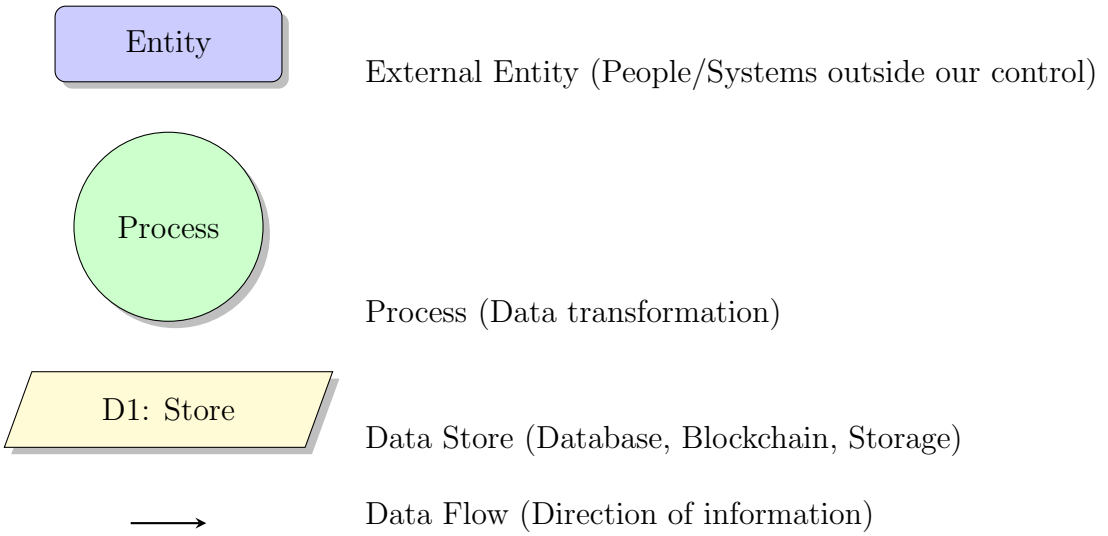
Visual representation of all data flows, processes, and interactions
in the Sizland Notary-Mediated Escrow System

January 23, 2026

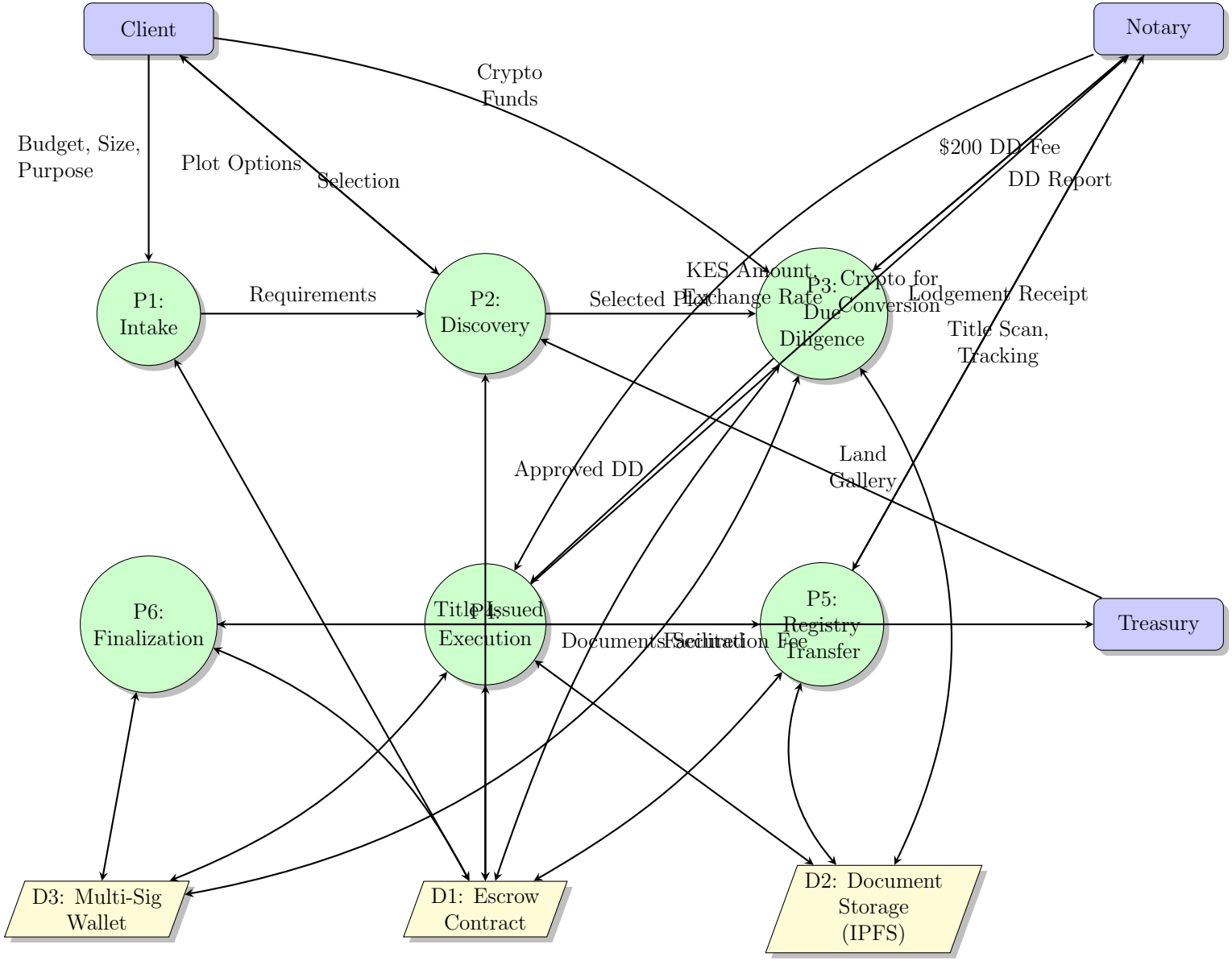
Data Flow Diagram - Level 0 (Context Diagram)



Legend



Data Flow Diagram - Level 1 (Detailed System)



Process Descriptions

P1: Intake Process

Input: Client requirements (budget, size, purpose), seller address, notary address
Process: Create intake record in escrow contract with all client specifications
Output: Escrow ID, state set to "Intake"
Data Stores Updated: D1 (Escrow Contract)

P2: Discovery Process

Input: Digital gallery from Sizland (geo-pins, photos, videos), client selection
Process: Display available plots matching criteria, capture client selection
Output: Selected plot details, trigger funding phase
Data Stores Updated: D1 (Escrow Contract with selected plot info)

P3: Due Diligence Process

Input: Client funding ($5000 or $200), notary verification work
Process: Release $200 to notary, conduct registry search and site visit, upload DD report
Output: DD Report document hash, verification status, client decision (Proceed/Pivot/Cancel)
Data Stores Updated: D1 (state change), D2 (DD report), D3 (multi-sig tx for $200 release)

P4: Execution Process (Fiat Bridge)

Input: Client approval, crypto funds in escrow
Process: Release crypto to notary, notary converts to KES, pays seller offline, secures documents
Output: KES amount paid, exchange rate, documents secured confirmation
Data Stores Updated: D1 (KES amounts, exchange rates, state), D2 (document hashes), D3 (multi-sig tx)

P5: Registry Transfer Process

Input: Secured documents, notary lodgement at registry
Process: Upload lodgement receipt, set expected completion date, track progress, upload title scan
Output: Lodgement receipt hash, progress updates, title document hash, courier tracking
Data Stores Updated: D1 (timeline, courier info), D2 (lodgement receipt, title scan)

P6: Finalization Process

Input: Title issued confirmation, courier tracking number
Process: Release facilitation fee to Treasury, mark escrow complete, ship physical title
Output: Completed escrow, facilitation fee transferred, tracking number
Data Stores Updated: D1 (state = Completed), D3 (multi-sig tx for fee release)

Data Store Specifications

D1: Escrow Contract (Blockchain)

Type: Ethereum Smart Contract (NotaryMediatedEscrow.sol)
Data Stored:

- Escrow ID
- Client, Seller, Notary addresses
- Budget, desired size, purpose
- Total amount, due diligence fee, deposit, final amount, facilitation fee
- Negotiated rates (Deposit
- KES amount paid to seller
- Exchange rate (crypto to KES)
- Escrow state (Intake, Discovery, Funded, etc.)
- Timestamps (created, funded, completed)
- Expected completion date
- Courier tracking number
- Documents secured flag

Access: Read (public), Write (authorized functions only)

D2: Document Storage (IPFS + On-Chain Hashes)

Type: Hybrid storage (IPFS for files, Blockchain for hashes)
Data Stored:

- Document hashes (SHA-256)
- Upload timestamps
- Uploaded by (address)
- Verification status
- Document type (Search Certificate, DD Report, Sale Agreement, Title, etc.)
- Signing status and signatures

Access: Hashes (public on-chain), Files (permissioned IPFS access)

D3: Multi-Sig Wallet (Blockchain)

Type: Ethereum Smart Contract (MultiSigWallet.sol)
Data Stored:

- Wallet owners (Doddy, Treasury, Third Party)
- Required confirmations (2 of 3)
- Transaction queue (pending, executed)
- Confirmation status per transaction per owner
- Transaction history with timestamps

Access: Read (public), Submit/Confirm/Execute (owners only)

Security and Data Flow Controls

- **Access Control:** All processes enforce role-based permissions (Client, Notary, Admin)
- **State Validation:** Each process validates the current escrow state before execution
- **Multi-Sig Protection:** All fund releases flow through the Multi-Sig Wallet (D3)
- **Document Integrity:** SHA-256 hashes ensure document authenticity
- **Audit Trail:** All state changes and transactions logged on-chain
- **Reversal Protection:** State machine prevents backward state transitions

This Data Flow Diagram represents the complete Sizland ecosystem showing all entities, processes, data stores, and information flows.